Lecture Notes

Lecture notes will appear after class when I have a chance to scan and post them.

- **Fri (https://q.utoronto.ca/courses/380078/files/37273559?wrap=1) (https://q.utoronto.ca/courses/380078/files/37273559/download?download_frd=1)** 4 Apr 2025 (last class!)
- **Wed (https://q.utoronto.ca/courses/380078/files/37212070?wrap=1) (https://q.utoronto.ca/courses/380078/files/37212070/download?download_frd=1)** 2 Apr 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/37152563?wrap=1) (https://q.utoronto.ca/courses/380078/files/37152563/download?download_frd=1)** 31 Mar 2025
- **Fri (https://q.utoronto.ca/courses/380078/files/37114406?wrap=1) (https://q.utoronto.ca/courses/380078/files/37114406/download?download_frd=1)** 28 Mar 2025 (we'll briefly discuss the plots on the last page on Mon)
- **Wed (https://q.utoronto.ca/courses/380078/files/37048063?wrap=1) (https://q.utoronto.ca/courses/380078/files/37048063/download?download_frd=1)** 26 Mar 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/36999378?wrap=1) (https://q.utoronto.ca/courses/380078/files/36999378/download?download_frd=1)** 24 Mar 2025 (we'll discuss the example on pgs 5+6 on Wed)
- **Fri (https://q.utoronto.ca/courses/380078/files/37077121?wrap=1) (https://q.utoronto.ca/courses/380078/files/37077121/download?download_frd=1)** 21 Mar 2025
- **Wed (https://q.utoronto.ca/courses/380078/files/36891375?wrap=1) (https://q.utoronto.ca/courses/380078/files/36891375/download?download_frd=1)** 19 Mar 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/36843716?wrap=1) (https://q.utoronto.ca/courses/380078/files/36843716/download?download_frd=1)** 17 Mar 2025
- **No lecture on Fri 14 Mar 2025 - class cancelled for this day**
- **Wed (https://q.utoronto.ca/courses/380078/files/36740954?wrap=1) (https://q.utoronto.ca/courses/380078/files/36740954/download?download_frd=1)** 12 Mar 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/36695281?wrap=1) (https://q.utoronto.ca/courses/380078/files/36695281/download?download_frd=1)** 10 Mar 2025

  - (optional) **Proof (https://q.utoronto.ca/courses/380078/files/36733828?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36733828/download?download_frd=1)** of the approximation error result.
- **Fri (https://q.utoronto.ca/courses/380078/files/36688026?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36688026/download?download_frd=1)** 7 Mar 2025
- **Wed (https://q.utoronto.ca/courses/380078/files/36688021?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36688021/download?download_frd=1)** 5 Mar 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/36557308?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36557308/download?download_frd=1)** 3 Mar 2025
- Fri 28 Feb 2025 ... **Midterm Test (https://q.utoronto.ca/courses/380078/pages/midterm-test-info-page)** day
- **Wed (https://q.utoronto.ca/courses/380078/files/36456921?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36456921/download?download_frd=1)** 26 Feb 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/36417238?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36417238/download?download_frd=1)** 24 Feb 2025
  - GE with Partial Pivoting **example (https://q.utoronto.ca/courses/380078/files/36395817?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36395817/download?download_frd=1)** (txt)
- **Fri (https://q.utoronto.ca/courses/380078/files/36335477?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36335477/download?download_frd=1)** 14 Feb 2025
- **Wed (https://q.utoronto.ca/courses/380078/files/36277074?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36277074/download?download_frd=1)** 12 Feb 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/36197260?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36197260/download?download_frd=1)** 10 Feb 2025
- **Fri (https://q.utoronto.ca/courses/380078/files/36165073?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36165073/download?download_frd=1)** 7 Feb 2025
- **Wed (https://q.utoronto.ca/courses/380078/files/36105435?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36105435/download?download_frd=1)** 5 Feb 2025
  - a **note (https://q.utoronto.ca/courses/380078/files/36106005?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36106005/download?download_frd=1)** on computing a matrix multiplication
- **Mon (https://q.utoronto.ca/courses/380078/files/36053173?wrap=1)** **(https://q.utoronto.ca/courses/380078/files/36053173/download?download_frd=1)** 3 Feb 2025

- **Fri (https://q.utoronto.ca/courses/380078/files/36001679?wrap=1) (https://q.utoronto.ca/courses/380078/files/36001679/download?download_frd=1)** 31 Jan 2025
  - a **worked example (https://q.utoronto.ca/courses/380078/files/36165378?wrap=1) (https://q.utoronto.ca/courses/380078/files/36165378/download?download_frd=1)** of a LU decomposition where interchanges are not needed
- **Wed (https://q.utoronto.ca/courses/380078/files/35934645?wrap=1) (https://q.utoronto.ca/courses/380078/files/35934645/download?download_frd=1)** 29 Jan 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/35895721?wrap=1) (https://q.utoronto.ca/courses/380078/files/35895721/download?download_frd=1)** 27 Jan 2025
- **Fri (https://q.utoronto.ca/courses/380078/files/35895730?wrap=1) (https://q.utoronto.ca/courses/380078/files/35895730/download?download_frd=1)** 24 Jan 2025
- **Wed (https://q.utoronto.ca/courses/380078/files/35808727?wrap=1) (https://q.utoronto.ca/courses/380078/files/35808727/download?download_frd=1)** 22 Jan 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/35752267?wrap=1) (https://q.utoronto.ca/courses/380078/files/35752267/download?download_frd=1)** 20 Jan 2025 (note there is an uncorrected typo in the date in the notes ...)
  - another **catastrophic cancellation example (https://q.utoronto.ca/courses/380078/files/35754377?wrap=1) (https://q.utoronto.ca/courses/380078/files/35754377/download?download_frd=1)** (and an explanation for issue from lecture 1 !)
  - an **example (https://q.utoronto.ca/courses/380078/files/35754094?wrap=1) (https://q.utoronto.ca/courses/380078/files/35754094/download?download_frd=1)** of decimal to binary conversion
- **Fri (https://q.utoronto.ca/courses/380078/files/35712095?wrap=1) (https://q.utoronto.ca/courses/380078/files/35712095/download?download_frd=1)** 17 Jan 2025
  - sample calculation **example (https://q.utoronto.ca/courses/380078/files/35704579?wrap=1) (https://q.utoronto.ca/courses/380078/files/35704579/download?download_frd=1)**
- **Wed (https://q.utoronto.ca/courses/380078/files/35681068?wrap=1) (https://q.utoronto.ca/courses/380078/files/35681068/download?download_frd=1)** 15 Jan 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/35655080?wrap=1) (https://q.utoronto.ca/courses/380078/files/35655080/download?download_frd=1)** 13 Jan 2025

  - Please start reading in the textbook. Read in the order section 1.1 and then section 1.3. Next week we'll discuss section 1.2 (yes, a bit out of order).
- **Fri (https://q.utoronto.ca/courses/380078/files/35606234?wrap=1) (https://q.utoronto.ca/courses/380078/files/35606234/download?download_frd=1)** 10 Jan 2025
  - Optional extra reading is now available on the **Optional Reading (https://q.utoronto.ca/courses/380078/pages/optional-reading)** page
- **Wed (https://q.utoronto.ca/courses/380078/files/35586151?wrap=1) (https://q.utoronto.ca/courses/380078/files/35586151/download?download_frd=1)** 08 Jan 2025
- **Mon (https://q.utoronto.ca/courses/380078/files/35541788?wrap=1) (https://q.utoronto.ca/courses/380078/files/35541788/download?download_frd=1)** 06 Jan 2025
  - Please review the syllabus.
  - Maclaurin series approximation for exp(x)
    - Python program (**py (https://q.utoronto.ca/courses/380078/files/35513945?wrap=1) (https://q.utoronto.ca/courses/380078/files/35513945/download?download_frd=1)**)
    - with results included (**pdf (https://q.utoronto.ca/courses/380078/files/35513943?wrap=1) (https://q.utoronto.ca/courses/380078/files/35513943/download?download_frd=1)**)
  - Undesirable display of floats (**pdf (https://q.utoronto.ca/courses/380078/files/35513942?wrap=1) (https://q.utoronto.ca/courses/380078/files/35513942/download?download_frd=1)**)